

The Constants of Nature

$E_{\rm h}:\text{K}=\frac{\text{kelvin}}{\text{K}}\left(\frac{T_{\rm p}}{m_{\rm p}}\right)\left(1-14\,C_{\rm D}^{-4}\,\mathscr{K}_{\Theta}\right)\boxtimes$	$m_{\rm de}=\frac{3s}{\mathscr{K}_5\,\mathscr{K}_4}\left(\frac{16}{4\pi}\right)\left(m_{\rm n}+m_{\rm +}\right)\left(1+\frac{Im(\omega_{\rm i})}{\mathscr{K}_2}\left(\frac{L_2}{14}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$\mathscr{E}_{\rm h}=\frac{\text{joule}}{\text{K}}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}^2}\right)\left(1-\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm h}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}\,m_{\rm e}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$c=\left(\frac{L_2}{T_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$F_{90}=\text{farad}\left(1+\frac{C_d}{6}\,(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4)\boxtimes\right)$	$K_j=\frac{\mathscr{K}_1}{\pi}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1+\mathscr{K}_2^2)\boxtimes\right)$	$g_{\rm de}=\frac{2\pi}{8\text{K}}\left(1+\frac{16}{t_{\rm p}^2}\,(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2)\boxtimes\right)$
$\text{K}:E_{\rm h}=\frac{\text{kelvin}}{\mathscr{K}_1^4}\left(\frac{m_{\rm p}}{t_{\rm p}\,m_{\rm e}}\right)\left(1+14\,C_{\rm D}^{-4}\,\mathscr{K}_{\Theta}\right)\boxtimes$	$m_{\rm u}=\frac{1}{\mathscr{K}_3\,\mathscr{K}_4}\sqrt{\frac{16}{4\pi}}\,(m_{\rm n}+m_{\rm +})\left(1+\left(\frac{16}{2\pi}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$R_{\rm K}=\frac{2\pi}{\mathscr{K}_1^2}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm N}}{hc}=\frac{\mathscr{K}_1}{4\pi}\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}\,m_{\rm e}}\right)\left(1+\sqrt{\frac{1}{D_{\rm D0}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{1}{\text{m}}:\text{Hz}=\frac{1}{\text{meter}}\left(\frac{1}{t_{\rm p}}\right)\left(1+\sqrt{G_{\rm G1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$H_{90}=\text{henry}\left(1-\frac{C_d}{6}\,(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4)\boxtimes\right)$	$\frac{e}{h}=\mathscr{K}_1\left(\frac{t_{\rm p}\,q_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1^2+\mathscr{K}_2^2)\boxtimes\right)$	$\frac{\mu_{\rm de}}{8\text{K}}\left(1+\frac{16}{t_{\rm p}^2}\,(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2)\boxtimes\right)$
$\Delta_{\text{vcc}}=\frac{E_1}{2\pi}\left(\frac{t_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$m_{\rm +}=\frac{-2\pi\mathscr{K}_1}{\mathscr{K}_3}\left(\frac{\text{GeV}}{\sqrt{5}}\right)\left(\frac{t_{\rm p}^2}{t_{\rm p}^2}\right)\left(1+\frac{1}{6^{1/2}}\left(\frac{2\pi}{2s}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$Z_0=4\pi\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$A_{90}=\text{ampere}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$r_{\rm e}=\mathscr{K}_1^2\left(\frac{t_{\rm p}}{m_{\rm e}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$\Omega_{90}=\text{ohm}\left(1-\frac{C_d}{6}\,(\mathscr{K}_1+\mathscr{K}_3+\mathscr{K}_4)\boxtimes\right)$	$\Phi_0=\frac{\pi}{\mathscr{K}_1}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1^2+\mathscr{K}_2^2)\boxtimes\right)$	$\frac{\mu_{\rm de}}{\mu_{\rm B}}=\frac{2\pi}{8\text{K}}\left(\frac{m_{\rm p}}{m_{\rm e}}\right)\left(1+\frac{16}{t_{\rm p}^2}\,(\mathscr{K}_1^2+\mathscr{K}_3^2+\mathscr{K}_4^2)\boxtimes\right)$
$\text{eV}:\text{Hz}=\frac{\text{eV}}{2\pi}\left(\frac{t_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$m_{\rm he}=\frac{\sqrt{3}}{\mathscr{K}_1^2\,\mathscr{K}_2^2}\left(\frac{\text{GeV}}{2\pi}\right)\left(\frac{t_{\rm p}^2}{t_{\rm p}^2}\right)\left(1-\frac{3^{-1}}{\mathscr{K}_1\,\mathscr{K}_2}\left(\frac{s}{18}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$Z_{\rm p}=\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}^2}\right)\left(1+\frac{1}{2}Re\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$C_{90}=\text{coulomb}\left(1+\sqrt{\frac{9}{J_{0.1}}}\,\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$a_0=\frac{1}{\mathscr{K}_1^2}\left(\frac{t_{\rm p}\,m_{\rm p}}{m_{\rm e}}\right)\left(1-\sqrt{6\,C_{\rm R1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$ST_0=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{t_{\rm p}^2\,T_{\rm p}\,m_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\,\mathscr{K}_2}\right)\,(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4)\boxtimes\right)$	$\frac{h}{e}=\frac{\text{coulomb}}{\mathscr{K}_1}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}\,q_{\rm p}}\right)\left(1+\frac{7}{8}\left(\frac{8}{4\pi}\right)^2\,(\mathscr{K}_1^2+\mathscr{K}_2^2)\boxtimes\right)$	$b^{\prime}=\frac{v_{\rm peak}}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{18}\left(\frac{2\pi}{s}\right)^2\,(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2)\boxtimes\right)$
$\text{J}:\text{Hz}=\frac{\text{joule}}{2\pi}\left(\frac{t_{\rm p}}{t_{\rm p}^2\,m_{\rm p}}\right)\left(1-\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$\frac{G}{hc}=\text{GeV}^2\left(\frac{t_{\rm p}^2}{t_{\rm p}^2\,m_{\rm p}}\right)^2\left(1-\left(\frac{4\pi}{18}\right)\,\mathscr{K}_{\gamma}{}^3\boxtimes\right)$	$\text{Hz}:\frac{1}{\text{m}}=\frac{1}{\text{second}}\left(\frac{t_{\rm p}}{t_{\rm p}}\right)\left(1+\frac{1}{2}Im\left(i^{i^{\prime\prime}}\right)\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\frac{\mu_{\rm h}^{'}}{S}=\frac{2\,\mathscr{K}_1\,\mathscr{K}_2^2}{S}\left(1-18\left(6\pi\right)\mathscr{K}_1\,\mathscr{K}_3\,\mathscr{K}_4\boxtimes\right)$	$\sigma_{\rm e}=\sqrt{\frac{32}{18}}\,2\pi\mathscr{K}_1^4\left(\frac{t_{\rm p}\,m_{\rm p}}{t_{\rm p}^2}\right)^2\left(1-2\sqrt{6\,C_{\rm R1}}\,(\mathscr{K}_1-\mathscr{K}_2)\boxtimes\right)$	$ST_1=\frac{5}{2}+\log\left(\left(\frac{1}{2\pi}\frac{\text{K}\,A_{\rm mass}}{t_{\rm p}^2\,T_{\rm p}\,m_{\rm p}}\right)^{3/2}\frac{\text{K}\,E_{\rm p}}{T_{\rm p}\,P_1}\right)\left(1-\frac{1}{12}\left(\frac{s^2}{\mathscr{K}_1\,\mathscr{K}_2}\right)\,(\mathscr{K}_1-\mathscr{K}_3-\mathscr{K}_4)\boxtimes\right)$	$c_1=(2\pi)^2\left(\frac{t_{\rm p}^4\,m_{\rm p}}{t_{\rm p}^2}\right)\left(1-\frac{4\pi^2}{32}\,(\mathscr{K}_1^2+\mathscr{K}_2^2)\boxtimes\right)$	$\mathscr{K}_B=\frac{1}{2\pi}\left(\frac{1}{t_{\rm p}\,T_{\rm p}}\right)\left(1+\frac{16}{16}\left(\frac{2\pi}{s}\right)^2\,(\mathscr{K}_1^2-\mathscr{K}_3^2-\mathscr{K}_4^2)\boxtimes\right)$
$\text{Hz}:\text{eV}=\frac{2\pi\text{joule}}{\text{second eV}}\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t_{\rm p}}\right)\left(1+\sqrt{\zeta(2)}\,\mathscr{K}_{\Theta}{}^2\boxtimes\right)$	$G=\left(\frac{t_{\rm p}^2\,m_{\rm p}}{t$						

$t_p = 5.39125836832313 \times 10^{-46}$ s	Planck time
$l_p = 1.61625918175645 \times 10^{-38}$ m	Planck length
$q_p = 1.87554596713962 \dots \times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795 \times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579 \times 10^{-8}$ kg	Planck mass
$\Xi = 0.999999199973626 \dots \times 10^{-7}$	hyperbolic universe boundary
5	number of dimensions
7	break in scale symmetry
$\pi_1 = 0.0854254531533304 \dots$	1 st hyperbolic partition constant
$\pi_2 = 3.66756753480501 \dots$	2 nd hyperbolic partition constant
$\pi_3 = 1.87649603900417 \dots + 0.46615262615972 \dots i$	3 rd hpc
$\pi_4 = 1.878649603900417 \dots - 0.46615262615972 \dots i$	4 th hpc
$\pi_5 = 4.47826244916751 \dots$	hyperbolic radius constant
$\pi_6 = 2.00316562310924 \dots$	hyperbolic radian constant
$G_{H_0} = 1.014944160640965 \dots$	Gieseking's constant
$V_k = 0.29598321281930 \dots$	figure eight knot hyperbolic volume
$K = 0.015965554772190 \dots$	Catalan's constant
$C_d = 0.291560904030818 \dots$	domino tiling constant
$D_4 = 1.79162281206959 \dots$	dimer constant
$\omega_1 = 0.7649747018528596 \dots + 1.3249706214087 \dots i$	omega_1 constant
$\omega_2 = 1.5295940370597192 \dots$	omega_2 constant
$i^{i^{i^{\cdot}}}$	power tower of i
$i = (-1)^{1/2}$	imaginary unit
$q_i = (-1)^{1/2}$	imaginary golden ratio
$G_g = 1.51872847301812 \dots$	tether length for grazing half the unit circle
$\pi = 3.14159265358979 \dots$	Archimedes' constant
$S = 2.29558714939263 \dots$	universal parabolic constant
$s = 5.244115108584261 \dots$	arc length of the unit lemniscate
$L = 2.622057554292119 \dots$	lemniscate constant
$L_1 = 1.31102877146205 \dots$	1 st lemniscate constant
$L_2 = 0.599070117367796 \dots$	2 nd lemniscate constant
$C_{ij} = 0.884713084793979 \dots$	ubiquitous constant
$G_{GK} = 0.834626841674073 \dots$	Gauss's constant
$C_{CF} = 0.697774657964819 \dots$	continued fraction constant
$R_{RR} = 0.9981360445908509 \dots$	Ramanujan's 1 st continued fraction constant
$F = 1.226742107202305 \dots$	Fibonacci factorial constant
$M = -1.477456459463318 \dots$	Modeling constant
$e = 2.71828182845904 \dots$	Euler's number
$\gamma = 0.577215664901532 \dots$	Euler-Mascheroni constant
$S = 0.82282549678847 \dots$	Sierpinski's constant
$J_{Bessel} = 2.4048255670595772 \dots$	1 st root of the Bessel function
$C_{FP1} = 1.19967864025773 \dots$	Real fixed point of the hyperbolic cotangent
$D_{N0} = 0.739085133215160 \dots$	Dottie number
$L_A = 0.662743419349181 \dots$	Laplace limit
$a_F = 2.50290787590589 \dots$	alpha Feigenbaum constant
$\delta = 4.66920160910299 \dots$	delta Feigenbaum constant
$S = 3.246479603071714 \dots$	silver constant
$p = 1.32471795724474 \dots$	plastic constant
$C_Q = 0.605443657196732 \dots$	QRS constant
$c_{QA} = 2.03816937970215 \dots$	associated QRS constant
$\chi_{ISM} = 0.3532363781584995 \dots$	Hafner-Sarnak-McCurley constant
$T_0 = 2.731500000000000 \dots \times 10^5$ kelvin	freezing point of water
$P = 1.000000000000000 \dots \times 10^2$ pascal	1 bar
$P_1 = 1.01325003754773 \dots \times 10^5$ pascal	standard atmospheric pressure
$E_1 = 6.09110229711386 \dots \times 10^{-24}$ joules	energy associated with Δv_{CS}
$f_1 = 5.400000000000000 \dots \times 10^{14}$ Hz	frequency of max light wavelength
$\lambda_{peak} = 4.96511423174427 \dots$	peak λ of spectral radiance/unit wavelength
$\nu_{peak} = 3.24149397212207 \dots$	peak emission of spectral flux/unit frequency
$\log(x)$ = natural logarithm	
$\Gamma(x)$ = gamma function	ln = lumen = cd · sr
$\zeta(x)$ = Riemann zeta function	N = newton = m kg/s ²
$\ln$ = derangement function	N = noether = m kg/s
	O = ohm = m ² kg/s ² C ²
units	P = alpha / kg/s ² m
m = meter	S = siemens = s C ² /m ² kg
m = second	T = tesla = kg/s C
C = coulomb	V = volt = m ² kg/s C
K = kelvin	W = watt = m ² kg/s ³
kg = kilogram	Wh = weber = m ² kg/s C
derived units	
MHz = megahertz = 1.000000000000000 ... × 10 ⁶ 1/s	
fm = femtometer = 1.000000000000000 ... × 10 ⁻¹⁵ m	
eV = electron-volt = 1.000000000000000 ... eV	
MeV = mega-electron-volt = 1.000000000000000 ... × 10 ⁶ eV	
GeV = giga-electron-volt = 1.000000000000000 ... × 10 ⁹ eV	
$r_e = \frac{4\pi}{(r_s)^2}$	$r_n = r_r = \sqrt{\frac{C_v}{4}}$
$r_o = \sqrt{\frac{2\pi}{\log(36)}}$	$\epsilon_p = \frac{l_p^2}{r_p^2} \frac{m_p}{\rho_p}$
external transform space	Physis Monastery Thad Roberts  October-27-2025
$V_{2k} = (\frac{4\pi}{5!}) (\frac{4\pi}{15}) (\frac{4\pi}{35}) (\frac{4\pi}{18}) (\frac{4\pi}{32}) (\frac{4\pi}{8})^4$	with: Angela Arvizu Vanessa Moiz Matthew Fox David Heggii Eschelon Azha Avi Rubin Wayne Eskridge Dan Whitaker Kevin Ritter Mike Sirois Jerry Garrard Shauna Montgomor
figure eight knot hyperbolic volume	Volfram Alpha NIST/CODATA
$V_8 = [Li_2(\frac{1}{\phi_1}) - Li_2(\phi_1)]$	
binomial constructor	
$A_{\text{ext } B_{\text{ext}}} (1 + A_{\text{int }} \Xi)$	$\Xi = \frac{l_p m_p}{\rho_p^2} \times \frac{\text{coulomb}^2}{\text{meter} \cdot \text{kilogram}}$
hyperbolic partition equation	$\alpha = \pi k^2$ $e = \pi k_j P$
$\frac{1}{x} + x$	